

Effect of mixing transition ions in glasses. II. The $\text{P}_2\text{O}_5\text{--Fe}_2\text{O}_3\text{--MnO}$ system

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Abstract

The first article of this series reported electrical properties of phosphate glasses containing V and Fe ions, both of which contributed to conductivity. A mixed transition-ion effect was detected in such glasses which had several characteristics similar to the mixed alkali effect, observed in glasses containing two alkali ions. This article reports electrical and optical properties of phosphate glasses containing Fe and Mn ions, one of which (Mn) does not contribute directly to conductivity. DC resistivity by the Van der Pauw four-probe method and optical absorption (UV–VIS–IR) measurements have been performed on $x\text{P}_2\text{O}_5 - (100 - x)(\text{Fe}_2\text{O}_3 + \text{MnO})$ (PFM) glasses. The transport mechanism has been identified to be hopping of small polarons between Fe^{2+} and Fe^{3+} sites. For $0.3 < n_{\text{Fe}} < 0.5$, where, n_{Fe} , the atomic fraction of Fe ions is given by $(\text{Fe}/(\text{P} + \text{Fe} + \text{Mn}))$, a marginal linear increase in resistivity has been observed with decreasing n_{Fe} . However, when n_{Fe} is reduced below ~ 0.3 , the resistivity increases exponentially with decreasing n_{Fe} , which is similar to ‘metal–insulator transition’ (MIT), observed in many crystalline systems. The resistivity transition in these glasses has been explained by a (small) polaron to (small) bipolaron transition (PBT) in the composition range $0 \leq n_{\text{Fe}} \leq 0.3$. The occurrence of PBT is confirmed by the shifting of the polaronic optical absorption band to higher energies with increasing small bipolaron concentration.

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1. Introduction

The first article in this series [1] reported a mixed transition ion effect (MTE) in $\text{P}_2\text{O}_5\text{--V}_2\text{O}_5\text{--Fe}_2\text{O}_3$ (PVF) glasses when the transition ion ratio (TIR), $(\text{V}/(\text{V} + \text{Fe}))$, was varied. Several characteristics of MTE were found similar to those of the mixed alkali effect (MAE), observed in many host glasses containing two alkali ions [2,3]. It was demonstrated that small polarons (SPs), located on both the V and Fe sites, contributed to the conduction process. The similarities between MAE and MTE are puzzling since the former is caused

by competitive transport of alkali ions (a classical barrier-crossing phenomenon), while the latter has its origin in small polaron (SP) conduction, a quantum process. One of the objectives of this investigation was to determine if a glass containing a different combination of TIs other than V and Fe would also exhibit MTE. Accordingly, the glass system chosen for this investigation was $\text{P}_2\text{O}_5\text{--Fe}_2\text{O}_3\text{--MnO}$ (PFM). However, as it will be demonstrated later, Mn ions do not participate in the conduction process and the exclusive carriers in PFM glasses are the SPs located on the Fe ions. MTE was not observed in PFM glasses since intermixing of SPs (self-trapped electrons contributed by different TIs) could not take place. However, the crucial role of MnO in the preparation of PFM glasses with n_{Fe} ($\text{Fe}/(\text{P} + \text{Fe} + \text{Mn})) < 0.4$ will be explained in a later section.

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Preparation of such glasses has made it possible to study the effect of electron correlation on the transport properties of PFM glasses.

Properties arising from the interaction between electrons or SPs are important in contemporary solid-state physics [4–13]. Several important phenomena such as superconductivity and giant/colossal magneto-resistivity (GMR/CMR) are explained by electron–electron correlation. In glasses, electron or SP-correlation may be studied by investigating electrical conduction and optical properties as the distance between the carriers is systematically varied, without changing the glass structure in a major way. In this investigation, such properties of PFM glasses have been studied as the Fe–Fe distance is varied by suitably changing the iron concentration.

Recently, Mansour et al. [7] and Doweidar et al. [8] have reported sharp increases in resistivity in TI-containing glasses (TI-glasses) at critical carrier concentrations, typical of the glass systems investigated. The origin of such resistivity transition has not been extensively addressed by the authors. Our investigation of the electrical and optical properties of phosphate glasses, containing mixtures of TIs such as Fe + Mn (present work) and V + Mn [9], indicate that such spectacular resistivity increase may be understood in the light of ‘metal–insulator transition’ as proposed by Mott [4].

The narrow energy band formed by the 3d electrons in TI-glasses, where each reduced TI contributes one electron, may be considered half-filled [13] since, according to the Pauli exclusion principle, each state can be occupied by two electrons with opposite spin directions. From the view point of Wilson’s [14] band structure of metals, the narrow band comprising the conduction electrons contributed by the reduced TIs resembles those for metals, except that the electrons are self-trapped, forming ‘Fermi glasses’ [10,11]. Under certain conditions, such a band may be energetically unstable and may split into two bands with an energy gap (Hubbard U) between them [4–6,10–13]. In such circumstances, the lower band will be completely filled (leaving the upper absolutely empty), resulting in an insulating state. As first shown by Mott [13,15,16] and described by Adler [5], correlation between carriers can lead to such a resistive transition, provided the distance between the carriers (in PFM glasses, this is the Fe^{2+} – Fe^{2+} distance) is greater than some critical value. Such materials are called Mott insulators. The structural parameters may be altered by the application of pressure or by changing the composition of a system so as to reach the insulating state from the metallic state by increasing the distance between the Fe ions. The transition from metallic to insulating behavior and vice versa, are known as Metal–Insulator (or Mott) Transition (MIT). In a system such as the PFM glasses, all the conduction electrons are self-trapped and are present as SPs. As every

Fe^{2+} contributes only one SP (out of the maximum two, with opposite spins) to every state, the band is half-full. Logically, if Mott transition takes place in such a material, the SPs should be converted into small bipolarons (SBPs).

Hubbard [17] introduced an important simplification into the problem of Mott transition, in an array of one-electron atoms, by including the interaction between a pair of electrons only if they are on the same atom. The model is an extension of the tight binding approach [10,11] and it restricts the overlapping of the electron orbitals only between adjacent charge centers. Though Hubbard’s original work dealt with s-electrons, according to Mott [4] and Adler [5], it is equally applicable to 3d electrons, which are the primary carriers of PFM glasses. These electrons may be considered to form a band of width B , for which Mott’s expression [11] gives:

$$B = 2zI, \quad (1)$$

where z is the number of nearest neighbors and I the overlap energy integral. If two electrons are on the same atom, their Coulomb repulsion contributes an energy, U , to the ensemble, comprising the charge carriers and the atom in consideration. As the distance between the atoms is varied, the band-width, B changes rapidly while U remains more or less constant if long-range repulsive interaction between carriers is ignored. If the distance between the centers is decreased, for example, U will change little but B will increase rapidly. When $B = U$, a metal–insulator transition occurs and it is sometimes referred to as the Mott–Hubbard transition.

MIT is one of the most vital and spectacular problems in contemporary condensed matter physics. Numerous theoretical and experimental studies have been conducted on this subject over the last several decades and these have been reviewed by Imada, Fujimori and Tokura [6]. The Mott criterion plays a crucial role in the development of the MIT theory [6,10–12] which states that:

$$a_{\text{H}} n_{\text{c}}^{1/3} \approx 0.25, \quad (2)$$

where a_{H} is the effective Bohr radius and n_{c} is the critical carrier density at the transition. This simple criterion provides a numerical prediction for MIT in many diverse systems such as doped semi-conductors, high- T_{c} superconductors, metal–ammonia solutions and metal–noble gas alloys. Unfortunately, there is no consistent and unified theory explaining MIT. Instead, there are a variety of models [6] such as the electron–correlation Mott–Hubbard transition, the Peirls transition in quasi one-dimensional systems with charge or spin-density waves, the Verwey transition (as in Fe_3O_4) involving charge ordering, the Anderson transition [18] due to disorder-induced localization, etc.

A literature survey on SP conduction in TI-glasses has been presented in the first article [1] of this series

and will not be repeated here. The basic equations for SP conduction are due to Austin and Mott [19] and Mott and Davis [11] and are given by

$$\sigma = \sigma_0 \exp(-W/kT), \quad (3)$$

where σ is the conductivity at high temperatures ($T > \theta/2$, θ being the Debye temperature), W is the activation energy for small polaron hopping (SPH) and σ_0 is given by

$$\sigma_0 = Nc(1-c)e^2vR^2/kT \exp(-2\alpha R), \quad (4)$$

where c is the redox ratio (reduced fraction) of TI participating in the conduction process, R is the distance between such TIs, e is the electronic charge, v is the longitudinal optical frequency, k is the Boltzmann constant, T is the absolute temperature and α is defined such that $\exp(-\alpha r)$ represents the wavefunction decay of a localized electron with distance, r . The activation energy for dc conduction is related to the energy involved in the formation of SPs (W_P) by [10,11]:

$$W \approx 1/2W_P. \quad (5)$$

As stated earlier, the present investigation was aimed at determining if PFM glasses would exhibit MTE as the PVF glasses did. Though MTE was not observed, a sharp increase in resistivity was observed at a critical Fe concentration exactly as Mansour et al. [7] and Doweidar et al. [8] observed in BaO–V₂O₅–B₂O₃ and Fe₂O₃–PbO–B₂O₃ glasses, respectively. The sharp increase in resistivity has been explained by a Mott–Hubbard type MIT caused by the transition of relatively mobile SPs to practically immobile SBPs at a critical carrier concentration.

2. Experimental

2.1. Glass preparation

Glasses in the series, $x\text{P}_2\text{O}_5 - (100 - x)(\text{Fe}_2\text{O}_3 + \text{MnO})$, were prepared from reagent grade P₂O₅, MnO and Fe₂O₃. The batches were melted in Pt/Au crucibles at 1473–1823 K for 30 min and quenched between thick copper plates. The glasses were not annealed after quenching and the absence of crystalline phases was confirmed by X-ray diffraction (XRD) analyses of glass powders. Glass compositions were analyzed by X-ray fluorescence (XRF) spectroscopy. Glass melting was performed in small batches (5 g) for achieving effective quenching, which was essential to avoid devitrification.

2.2. Resistivity measurements

Resistivity (ρ) was measured by the Van der Pauw four-probe method as described previously [1]. All electrical measurements were performed under 8–10 milli-

Torr vacuum, in a dark chamber. The uncertainty in ρ measurements has been estimated at $\pm 7.5\%$. The error in the determination of W is estimated at ± 0.03 eV.

2.3. Density measurements

Density measurements were carried out in standard pycnometers using deionized water and applying temperature corrections where necessary. The error in density measurements has been estimated at ± 0.01 g/cm³.

2.4. Redox equilibria

Redox ratios, ($\text{Fe}^{2+}/(\text{Total Fe})$), of iron (c) were determined by using a Mössbauer Spectrometer model MS-1200 (Ranger Scientific, Oak Ridge, TN). The error in the reported values of c is estimated at $\pm 20\%$.

2.5. Optical absorption spectroscopy

Optical absorption was measured in a dual-beam Spectrophotometer (Perkin–Elmer, Model 330) capable of scanning between 185 and 2600 nm. Samples were glass wafers polished to a thickness of around 200 μm and scanned at the rate of 20 nm per minute. The absorption was measured on an arbitrary scale since the objective was to determine the location of the absorption bands and not their relative intensities.

3. Results

Glass compositions, listed in Table 1, have been renormalized ignoring minor impurities which ranged between 0.5 and 1.0 wt%. The ρ at 500 K and the activation energy for dc conductivity (W) of the glasses are listed in Table 1. For the purpose of comparing transport properties, W was calculated from the slope of log conductivity (σ) vs $1/T$ lines (Arrhenius plots) in the temperature range 470–530 K for every sample (Fig. 1). As the slopes of the Arrhenius plots decreased monotonically with decreasing temperature (inset of Fig. 2), a short enough temperature range was chosen for the determination of W where such plots maintained reasonable linearity. W remains nearly invariant at $\sim 0.64 \pm 0.04$ eV and is independent of the composition of the glass, Table 1. Similar results were reported by Santic et al. [21] and by several other authors in the references cited therein. The pre-exponential factor, σ_0 , was calculated from the intercept of such plots and it varies between 0.11 and 30.72 (Ωcm)^{−1}. A marginal increase in ρ is detected as n_{Fe} decreases from ~ 0.5 to ~ 0.3 (Fig. 2). In the composition range $0 \leq n_{\text{Fe}} \leq 0.3$, ρ increases exponentially with decreasing n_{Fe} and goes beyond the detection limit of our apparatus at $n_{\text{Fe}} \approx 0.05$. The resistivity

Table 1
Composition and transport data of PFM glasses

Glass ID	Wt% (XRF analysis)			Atomic fraction of iron (n_{Fe})	ρ (at 500 K)	σ_0 (Ωcm) ⁻¹	W (eV)
	P ₂ O ₅	Fe ₂ O ₃	MnO				
PF-1	46.51	51.93	0.01	0.50	7.57E+04	13.93	0.60
PFM-1	45.92	49.48	3.40	0.47	2.28E+05	18.95	0.66
PFM-2	48.37	44.49	5.60	0.42	1.82E+05	30.72	0.67
PFM-3	45.33	42.46	10.29	0.40	2.90E+05	8.06	0.63
PFM-4	46.21	41.60	11.18	0.39	2.37E+05	16.90	0.66
PFM-5	42.11	42.79	14.02	0.40	9.69E+04	12.85	0.60
PFM-6	45.36	36.91	16.97	0.34	3.79E+05	10.19	0.65
PFM-7	45.22	33.10	20.72	0.31	5.14E+05	6.18	0.64
PFM-8	50.00	29.00	21.00	0.27	7.84E+05	4.10	0.65
PFM-9	45.13	28.85	25.58	0.27	1.00E+06	4.31	0.66
PFM-10	43.65	27.80	27.56	0.26	1.67E+06	2.25	0.65
PFM-11	38.16	22.23	38.10	0.21	1.81E+06	2.33	0.66
PFM-12	56.53	16.67	26.16	0.15	7.57E+07	0.11	0.68
PFM-14 ^a	55.07	5.95	38.38	0.05	6.13E+08	–	–

– Values not calculated.

^a Data extrapolated from high temperature run.

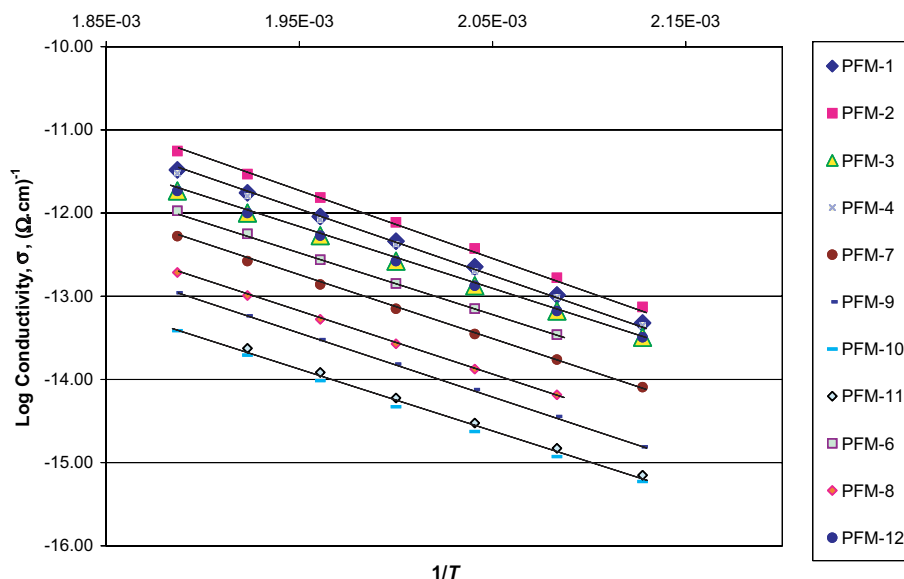


Fig. 1. Log conductivity vs $1/T$ plot of PFM glasses.

of the glass with $n_{\text{Fe}} \approx 0.05$ in (Table 1 and Fig. 2) is the extrapolated value obtained from a higher temperature run between 650 and 700 K which was necessary because sample resistance was higher than the detection limit of the equipment at 530 K. The large resistivity increase in a relatively short range of n_{Fe} has the features of metal–insulator transition (MIT).

Table 2 lists the average distance between the Fe ions, R ($N^{-1/3}$) [3,8], where N is the concentration of Fe ions in unit volume of glass which is determined from density measurements and assuming homogeneous distribution of iron in the glass. The polaron radius, r_p , is given by [7,8,22–25]:

$$r_p = 1/2(\pi/6N)^{1/3}. \quad (6)$$

The experimentally determined values of the redox ratio, redox factor, Fe–Fe distance (R), SP radius, r_p and Fe concentration (N) of the PFM glasses are provided in Table 2. Substituting $1/R^3$ for N in Eq. (4) gives:

$$\sigma_0 = c(1-c)e^2 v / RkT \exp(-2\alpha R). \quad (7)$$

The redox ratio (c) and $(c(1-c))$, the redox factor, determined from Mossbauer measurements, are listed in Table 2. As the random variation of the redox factor (between 0.19 and 0.23) falls within the error limits of Mossbauer measurements, $(c(1-c))$ is considered constant. Assuming v also to be a constant, Eq. (7) may be rewritten as:

$$\ln(R \cdot \sigma_0) = \ln E - 2\alpha R, \quad (8)$$

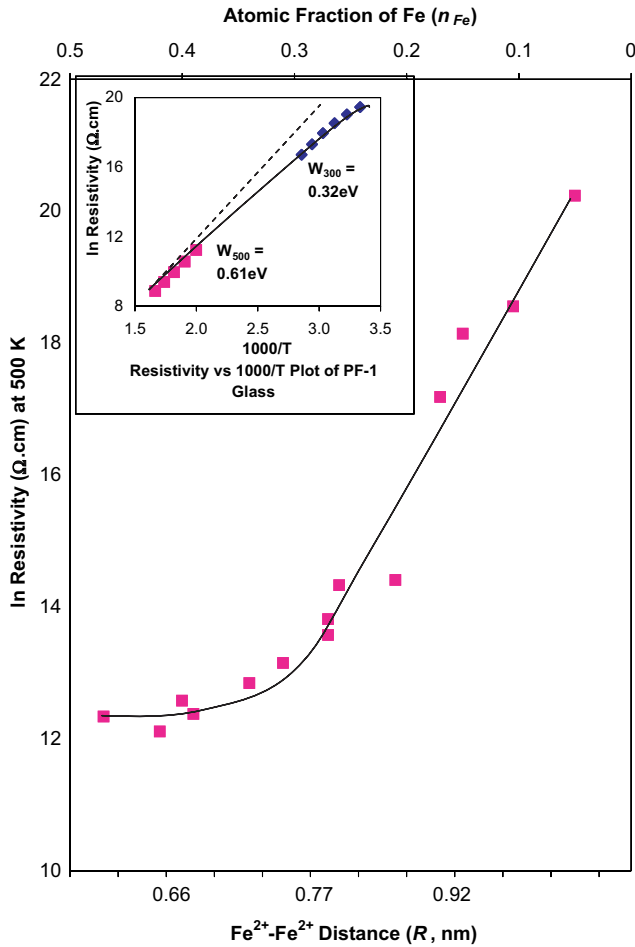


Fig. 2. Variation of resistivity with Fe–Fe distance and iron concentration W_P at 300 K = $0.32 \times 2 = 0.64$ eV. Therefore the optical absorption band is expected to be located at $2 \times 0.64 \approx 1.3$ eV, which roughly corresponds to 1000 nm.

where $E = c(1 - c)e^2v/kT$. A plot of $\ln(R \cdot \sigma_0)$ vs R should give a straight line and Mansour et al. [7] and

Doweidar et al. [8] have calculated the value of α in various host glasses (containing Fe ions) from the slopes of such plots. The value of α was determined to be $\sim 7.0 \text{ nm}^{-1}$, regardless of iron concentration or the composition of the host glass. α , calculated from the slope of $\ln(R \cdot \sigma_0)$ vs R of the PFM glasses (Fig. 5), gives a value of 7.3 nm^{-1} in the composition range $0.3 < n_{\text{Fe}} < 0.5$. Below $n_{\text{Fe}} = 0.3$, the plot of $\ln(R \cdot \sigma_0)$ vs R is observed to deviate from linearity, indicating that Eq. (8) is not valid in the composition range, $0.0 < n_{\text{Fe}} < 0.3$.

The optical absorption spectra of one manganese phosphate (PM) and an iron phosphate (PF) glass along with three PFM glasses in the composition range $0.00 \leq n_{\text{Fe}} \leq 0.45$ are illustrated in Fig. 4. The spectra (smoothened for better illustration) comprise broad and noisy absorption bands, the peak positions of which are listed in Table 2 along with the corresponding glass compositions. The absorption band at 1050 nm (for the PFM-14 glass) is observed to move to higher wavelengths with increasing n_{Fe} . The corresponding absorption band for the PF-2 glass ($n_{\text{Fe}} = 0.45$) is located at 1430 nm.

4. Discussion

According to Mott [10] and Mott and Davis [11] some of the characteristics of SP-hopping are: (i) low values of σ_0 , compared to the conductivities of normal metals, (ii) decreasing activation energy with decreasing temperature and (iii) thermopower activation energy lower than that for dc conduction. The pre-exponential co-efficient of the PFM glasses has relatively small values (Table 1) and the slopes of $\log \sigma$ vs $1/T$ plots of all the TI-glasses have been found to decrease continuously with temperature, as illustrated in the inset of Fig. 2 for the PF-1 glass. The activation energy calculated from

Table 2
Redox ratio of iron and optical absorption data of PF and PFM glasses

Glass ID	Iron concentration $N [\text{cm}^{-3}]$	Atomic fraction of iron [$n_{\text{Fe}} = \text{Fe}/$ (Fe + Mn + P)]	Redox ratio, c , [$\text{Fe}^{2+}/(\text{Total Fe})$]	R (nm)		$NR^2 \text{exp}$ ($-2\alpha R$)	r_P (nm) $1/2(\pi/6N)^{1/3}$	SP band wavelength, [nm]
				(Fe–Fe dist.)	(Fe ²⁺ –Fe ²⁺ dist.)			
PF-2	–	0.45	–	–	–	–	–	1430
PFM-1	1.08E+22	0.47	0.32 (0.22) ^a	0.45	0.66	3.07E+18	0.18	–
PFM-2	1.11E+22	0.42	0.29 (0.21)	0.45	0.68	3.16E+18	0.18	1380
PFM-3	8.15E+21	0.40	0.50 (0.25)	0.50	0.63	1.38E+18	0.20	1350
PFM-4	9.98E+21	0.39	0.28 (0.20)	0.46	0.71	2.56E+18	0.19	1330
PFM-5	8.73E+21	0.40	0.27 (0.19)	0.49	0.75	1.64E+18	0.20	1250
PFM-6	9.06E+21	0.34	0.29 (0.20)	0.48	0.72	1.89E+18	0.19	1150
PFM-7	8.09E+21	0.31	0.27 (0.20)	0.48	0.77	1.69E+18	0.20	1150
PFM-9	7.01E+21	0.27	0.31 (0.22)	0.52	0.77	9.56E+17	0.21	1100
PFM-10	6.90E+21	0.26	0.30 (0.21)	0.53	0.78	8.45E+17	0.21	1100
PFM-11	5.62E+21	0.21	0.26 (0.19)	0.56	0.88	4.96E+17	0.23	–
PFM-12	3.62E+21	0.15	0.35 (0.23)	0.65	0.92	1.16E+17	0.26	–
PFM-14	–	0.05	–	–	–	–	–	1050

– Not determined.

^a Values of redox factor, $c(1 - c)$, in parentheses.

thermopower measurements [20] has been found to be at least one order of magnitude lower than that determined from conductivity measurements. Moreover, the criterion for SPH conduction used by Doweidar et al. [8] and Gohar et al. [22,23], $1/\alpha < r_P < R_{\text{Fe-Fe}}$ ($0.14 < 0.18 < 0.45$) is satisfied for all the PFM glasses of this investigation (Table 2), confirming the prevalence of SPH transport.

To determine the role of Mn on conductivity, a few glasses were prepared by substituting Mn with P in PFM glasses containing a range of iron concentrations. These are the binary iron phosphate glasses in Fig. 3, the properties of which are compared with PFM compositions with comparable n_{Fe} . The ρ and W data indicate that glasses with and without Mn ions have very similar transport characteristics. However, when Fe_2O_3 is replaced by equimolar P_2O_5 (leaving the MnO concentration unchanged) the resistivities exceed the detection limit of the equipment used in this investigation. Hence, it is inferred that Fe ions only are responsible for dc conduction and Mn ions do not have a direct bearing on dc conductivity at the temperatures employed in the present investigation. At low temperatures ($T < \theta/2$), however, Mn ions may have significant influence on variable range [11,19] or nearest neighbor hopping [10,11] by influencing the itinerant carriers via magnetic interactions which are likely to become active at such low temperatures. In our discussion, hereafter, Mn ions will be treated as inert fillers as far as electrical properties are concerned. However, the importance of MnO during the preparation of glasses in the composition range $0.05 \leq n_{\text{Fe}} \leq 0.45$ will be discussed in a subsequent section. In the following, we shall critically assess the contribution of all the variables appearing in Eqs. (3) and (4) and attempt to explain the variation of resistivity, and especially the apparent observation of MIT, as a function of Fe concentration in the PFM glasses.

The redox factor (Table 2) is found to vary between 0.19 and 0.23, which is within the error limits of the Mossbauer data. Hence, we infer that variation of redox ratio of Fe ions, as a function n_{Fe} , cannot be held responsible for any variation in resistivity of the PFM glasses. The variables which are most likely to affect conductivity are W , R and α , because these parameters appear as $\exp(-W/kT)$ and $\exp(-2\alpha R)$ in Eqs. (3) and (4). The activation energy of the binary PF-1 glass is somewhat lower than that of the ternary PFM glasses (Table 1) but the scatter among W of the PFM glasses is barely above the error limit (± 0.03 eV). Moreover, the variation in W does not follow any pattern with changes in n_{Fe} . The small scatter in W appears to have been introduced by minor impurities which were found in composition determination by XRF. Similar results on the near-invariance of W with composition were reported by Santic et al. [21] and Gohar et al. [22,23] in phosphate and vanadate glasses containing wide ranges of iron concentration. Hence, it is reasonable to rule out W as the possible source of any variation of resistivity in the PFM glasses.

Doweidar et al. [8] and Gohar et al. [22,23] have determined that α remains nearly constant in phosphate and borate glasses with large variations in iron concentration. As the wavefunction decay parameter is determined by the size and shape of the potential well of a localized electron, its value should not be determined by the concentration of such electrons. As such its value is given by $\alpha = (2mW_0)^{1/2}/\hbar$ [11], where m is the effective mass of the electron and W_0 is the depth of an isolated potential well. If the effective mass of the electrons remains constant in these glasses, α should be unaffected by n_{Fe} . In the case of $\text{B}_2\text{O}_3\text{--Fe}_2\text{O}_3\text{--PbO}$ glasses, Doweidar et al. [8] has demonstrated that α (calculated from the slope of the $\ln(\sigma_0 \cdot R)$ vs R plot remains constant ($\sim 7.3 \text{ nm}^{-1}$) despite large variations in Fe concentra-

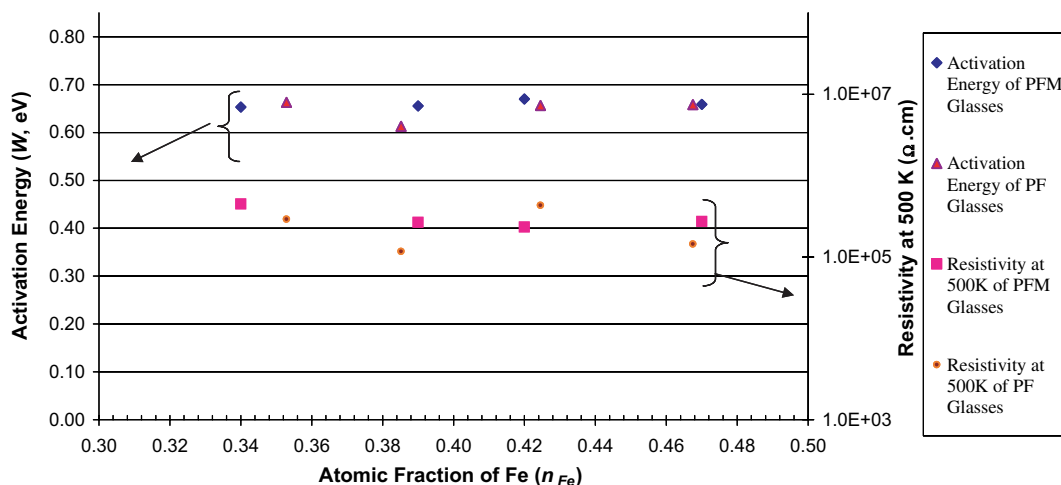


Fig. 3. Comparison of resistivity and activation energy of PF and PFM glasses with similar iron concentration.

tion. Gohar et al. [9] found α of V_2O_5 – Fe_2O_3 – BaO glasses to vary between 6.4 and 7.8 nm⁻¹, the average of which (7.1 nm⁻¹) is very close to the value obtained by Doweidar et al. [8] even though the glass systems were entirely different. For the PFM glasses of the present investigation, α , calculated from the linear portion of the $\ln(\sigma_0 \cdot R)$ vs R plot varies between 7.0 and 7.6 nm⁻¹, the average of which (7.3 nm⁻¹) will be used during the following discussion.

If the optical frequency of the PFM glasses is assumed constant, the variation in resistivity should be explained only by a corresponding variation of the term $[NR^2 \exp(-2\alpha R)]$. When n_{Fe} is decreased from 0.47 to 0.34, the resistivity increases by a factor of 1.66 and the reciprocal of $[NR^2 \exp(-2\alpha R)]$ increases by a factor of 1.09 (Table 2). As W , α and the redox factor ($c(1 - c)$) are invariant, the linear increase in resistivity with decreasing n_{Fe} may reasonably be described by the decrease in the concentration of SPs and the associated variation in R , the distance between Fe ions.

In the composition range $0.15 \leq n_{Fe} \leq 0.3$, however, the resistivity varies by a factor of $\sim 2 \times 10^2$ when the reciprocal of $[NR^2 c(1 - c) \exp(-2\alpha R)]$ varies by a factor of only 1.76. Eq. (8) is not valid in this composition range, where $\ln(R \cdot \sigma_0)$ does not vary linearly with R . It follows that a conventional description of SPH conduction, cannot explain the exponential increase in resistivity in this composition range. Fig. 5 clearly indicates that the actual values of R in the composition range, $0.15 \leq n_{Fe} \leq 0.3$ may be different from those calculated from the chemistry and density of these glasses. In the following, we shall attempt to interpret the unexpected increase in resistivity (apparent MIT) by the correlation effect among small polarons. In this process we shall utilize established theories on electron–electron [4,5,13,15,16] and SP–SP interaction in crystalline and disordered systems [26–31].

One method of studying the effect of SP-correlation in a non-crystalline material is to measure conductivity as a function of spatial distance between the carriers [3,13,20]. Binary iron phosphate (PF) glass could be ideal for such a study because of its ability to incorporate large quantities of Fe (up to 50 mol.%) and because a one-to-one correlation of conductivity with Fe–Fe distance is possible. However, during the melting of PF glasses in the composition range $0 \leq n_{Fe} \leq 0.4$ (where most correlation effects are observed), a portion of the P_2O_5 boils off, leaving the glass richer in Fe than intended. However, we have observed that for $n_{TI} \geq 0.4$, P_2O_5 loss is negligible where n_{TI} is the concentration of a mixture of TIs, such as Fe + Mn. By keeping the total TI concentration (n_{Fe+Mn}) at ~ 0.4 , P_2O_5 boil-off has been minimized and glasses with iron concentration in the range $0 \leq n_{Fe} \leq 0.4$ has been obtained. In other words, a series of glasses with n_{Fe} ranging between 0 and 0.4 could be prepared in atmospheric conditions

only by using MnO as a filler material such that n_{Fe+Mn} would always be ~ 0.4 . As Mn ions proved to be electrically inert, investigation of conductivity as a function of Fe–Fe distance became possible.

To our knowledge, the incidence of a Mott–Hubbard type ‘metal–insulator transition’ (MIT) in melt-processed oxide glasses has not been reported in the literature. The relevance of the Mott criterion (Eq. (2)) to non-crystalline lattice of melt-processed glasses, where all the charges are self-trapped, has been undertaken in the third article of this series [9]. The single-polaron problem, however, has been extensively studied [12,24,26] and a comprehensive understanding of polaron-physics has emerged. However, theories dealing with correlation between polarons in a multi-polaron scenario are rare. Recently, Capone and Ciuchi [27] have shown that strong correlation between SPs may lead to MIT following the arguments laid out in the half-filled Holstein model [12,28]. Meyer et al. [29] investigated electron–phonon (EP) coupling in a multi-SP scenario by the dynamical mean-field theory (DMFT) and predicted the formation of a gap in the density of states in the Holstein model, signaling a transition to the insulating state. These recent studies conclude that correlations between polarons may, indeed, lead to an insulating state via bipolaron formation just as correlation between electrons creates the Mott or Mott–Hubbard insulators. It should be noted, however, that these models are all based on crystalline lattice and do not consider the effect of Anderson localization [18], which is ubiquitous in all melt-processed glasses. Only recently, Bronold et al. [30] have studied Anderson localization in disordered EP systems in a single-polaron scenario and theoretical results of the latter also indicate the incidence of MIT at a critical SP concentration.

In the following, it is shown that, at a critical R_{Fe-Fe} , two neighboring SPs may undergo a Mott–Hubbard type transition to form a small bipolaron (SBP). Alexandrov and Mott [12] and Emin [31] have theoretically demonstrated that mass divergence may result when two electrons (with opposite spin) are localized in the same potential well. As a result, the SBP formed may be immobile for all practical purposes, causing an MIT-type transition. In crystalline materials, the transition is much sharper than that observed in PFM glasses. The relatively broad transition in glasses is not unexpected since the Fe ions are randomly dispersed in the glass and R_{Fe-Fe} actually has a broad range of values.

Emin [31] has described the optical properties of SPs and SBPs and developed a formalism governing the transformation from one to the other. An Emin-SBP comprises two electrons occupying the same potential well, the stability of which derives from a negative effective correlation energy despite the inherent coulomb repulsion between the electrons. Emin’s derivations of

the energies involved in the formation of SP (E_{SP}) and SBP (E_{SBP}) have established that bipolarons may be energetically stable with respect to separation into two polarons if the static dielectric constant ϵ_0 and W_P are sufficiently large such that $E_{SBP} < 2E_{SP}$. It was shown that, under favorable conditions, a second carrier within a single potential well quadruples the EP coupling energy, which facilitates SBP formation. It was also shown that, for an SBP, the depth of the potential well is twice as deep as that for an SP; $-2W_P$ rather than $-W_P$. The electronic energy of the two self-trapped carriers is $-2(2W_P) + U$, where U is the on-site Coulomb repulsion energy. An SBP will be more stable than an SP if $-2W_P + U > -W_P$, if Coulomb repulsion between carriers on adjacent sites is ignored. In other words, from energy considerations, W_P must be larger than U for SPs to be transformed into SBPs.

Following Emin's model [31], SP optical absorption is described by the excitation of a self-trapped carrier to an adjacent localized site which is energetically similar. For carrier transport between such a pair of sites, the energy involved is roughly twice the SP formation energy, $-2W_P$. The Emin-model of SP and SBP predict that the effective mass of the latter is considerably higher than the former resulting in substantially reduced mobility of the heavier pseudo-particle. Moreover, the optical absorption bands ascribed to SBPs have been shown to occur at higher energies than those due to SPs.

The characteristic absorption band of the PF and PFM glasses, located in the range 1050–1430 nm, is clearly attributable to iron ions because it is absent in the binary PM-glass. The general features of the SP absorption band of PFM glasses are that they are broad and asymmetric, which are characteristics of small polaron bands [12,31] which we observe in the absorption spectra of the PF/PFM glasses (Fig. 4). Absorption by a small polaron involves the difference in the electronic energies between two sites, which is just the lowering of the electronic energy associated with its formation. The energy involved is, roughly, double the energy for polaron formation (W_P) which is related to hopping conduction via Eq. (5). According to Emin's model, the polaron optical band is expected to be roughly around 1000 nm which is the wavelength corresponding to twice the W_P and four times the activation energy for dc conduction at 300 K (1.3 eV, inset of Fig. 2). Here, the activation energy at 300 K has been considered because optical absorption measurements were carried out at room temperature (~ 300 K). In these glasses, only one optical absorption band is observed in the 1000–1500 nm range. Hence, it is reasonable to attribute it to polaronic activity.

The absorption peak of the PF-2 glass ($n_{Fe} = 0.45$) is located at 1430 nm whereas that of the PFM-14 glass ($n_{Fe} = 0.05$) is at 1050 nm (Fig. 4). In other words, the absorption peak moves to higher energy as the Fe–Fe

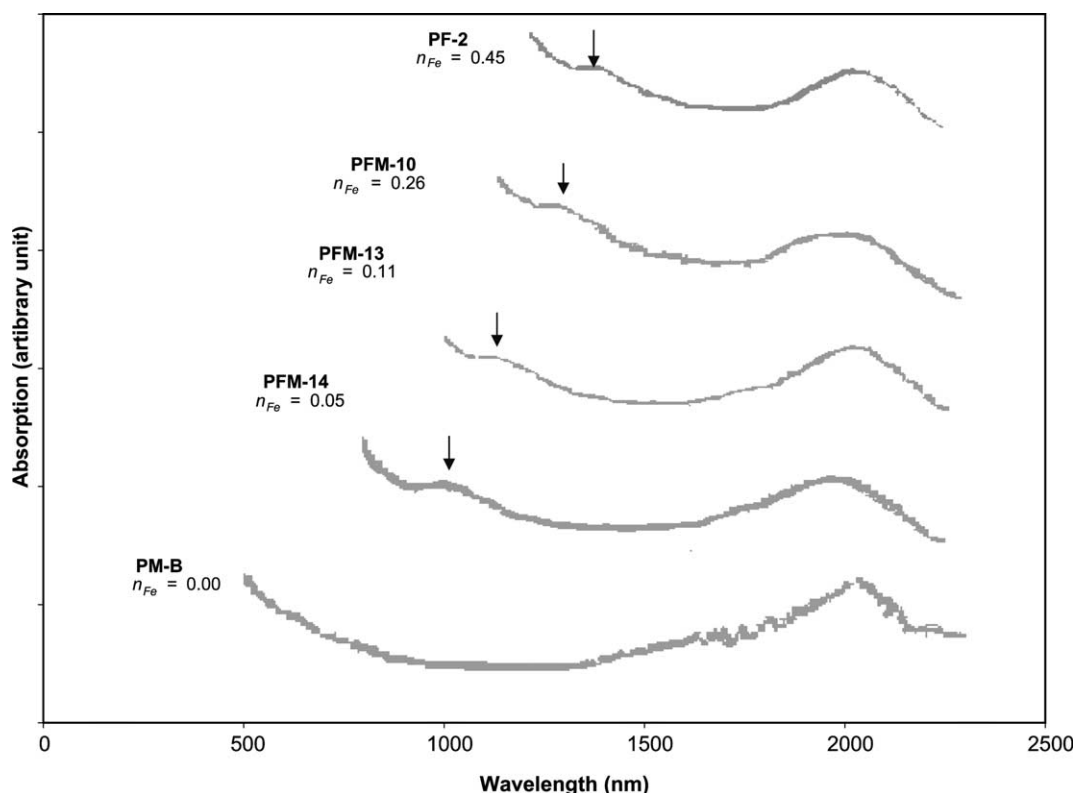


Fig. 4. Smoothed optical absorption spectra of PM-B and PFM glasses.

distance increases with decreasing iron concentration. Following Emin's suggestion this peak-shift may be explained by the occurrence of a small (polaron) to small (bipolaron) transition (PBT). At $n_{\text{Fe}} > 0.3$, the carriers are mostly small polarons and hence the absorption bands are located at relatively low energies. At $n_{\text{Fe}} < 0.3$, SBPs begin to appear in increasing numbers and so the absorption band moves to increasingly higher energies. Hence, the resistivity transition observed in these glasses may be explained to be a direct consequence of a PBT. Capone and Ciuchi's model [27] and especially the theoretical demonstration of Bronold et al. [30] regarding polaron to bipolaron crossover in Anderson localized materials seems to be verified by the experimental data of the present investigation. Further evidence of metal–insulator transition in oxide glasses will be provided in Part III of this series of articles [9], where a similar exponential resistivity increase in $\text{P}_2\text{O}_5\text{--V}_2\text{O}_5\text{--MnO}$ glasses, at a critical concentration of vanadium ions, is reported.

One important feature of the PFM glasses is the near-invariance of the activation energy with respect to composition (Tables 1, 2 and Fig. 3). The explanation for such behavior is provided in the following. For glasses with $n_{\text{Fe}} > 0.3$, most of the carriers are present as SPs, which increasingly transform into SBPs in the region $n_{\text{Fe}} < 0.3$. Though SPs and SBPs are present in the PFM glasses for all values of n_{Fe} , the ratio of the former to the latter decreases drastically as n_{Fe} is reduced below 0.3.

As the SBPs are essentially immobile, the hopping of the SPs is the only contribution to the conduction process. Hence the activation energy is actually the SP-hopping energy, which varies little despite large changes in n_{Fe} . The resistivity, however, increases exponentially in the region $0.15 \leq n_{\text{Fe}} \leq 0.3$ because the concentration of the small polarons rapidly decreases as a result of the inception of PBT. The immobile characteristic of SBPs has been reported by Alexandrov and Bratkovsky [32] in doped manganites where strong EP interaction results in a PBT-type conversion to a highly insulating state. On account of PBT, the distance between the SPs in PFM glasses increase but is not reflected in the values determined from the chemistry and the density of glasses. Consequently, the linearity of the $\ln(R \cdot \sigma_0)$ vs R plot is violated (Fig. 5).

One argument against the occurrence of PBT might be the invariance of the redox ratio with iron concentration. When two electrons are localized around an iron ion, c is expected to vary. The explanation lies in the atomistic description of SPs in TI-glasses proposed by Mott [24] and Greaves [25] which envisages the SP in these glasses to be an electron, trapped in an oxygen vacancy near a Fe^{3+} ion (converting it to a Fe^{2+} site). An SBP may be envisaged as two such Fe^{3+} ions around a neutral oxygen vacancy with one electron available

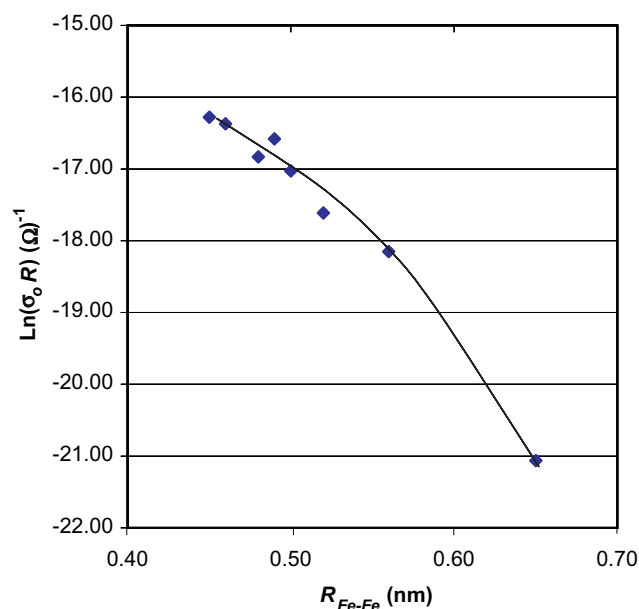


Fig. 5. $\ln(\sigma_0 R)$ as a Function of $R_{\text{Fe-Fe}}$.

for each of the two Fe^{3+} ions. In effect, two Fe^{3+} ions and two electrons belonging to the oxygen vacancy comprise an SBP. The two electrons (with opposite spins) may be considered to be occupying the same potential well (trap) while maintaining a constant redox ratio. One obvious method to test our hypothesis of the small polaron to small bipolaron transition or PBT in PFM glasses would be to study diamagnetic susceptibility response of these glasses as a function of Fe concentration.

5. Conclusion

DC resistivity and optical absorption measurements on $x\text{P}_2\text{O}_5 - (100 - x)(\text{Fe}_2\text{O}_3 + \text{MnO})$ glasses have shown evidence of a rapid transition from small polaronic conductivity to a highly insulating state on account of the formation of small bipolarons. Resistivity data reveal that manganese does not have any direct influence on the electrical properties of these glasses. The conduction process can be described by the hopping of small polarons between sites occupied by iron ions having different oxidation states. While the activation energy for dc conduction remains practically invariant in all the glasses, the resistivity increases marginally when the iron concentration ($\text{Fe}/(\text{Fe} + \text{Mn} + \text{P}) = n_{\text{Fe}}$) is reduced from ~ 0.5 to ~ 0.3 . However, below $n_{\text{Fe}} \approx 0.3$ it increases rapidly with decreasing iron concentration, exhibiting a metal–insulator type transition. At a critical Fe–Fe distance, which corresponds to $n_{\text{Fe}} \approx 0.3$, small polarons transform into small bipolarons. The latter, being much heavier than the former, are essentially

immobile and do not contribute to the conduction process. The (small) polaron to small (bipolaron) transition (PBT) is confirmed by the shifting of the small polaron optical absorption band to higher energies as the bipolaron concentration increases.

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References

- [1] B. Dutta, N.A. Fahmy, I.L. Pegg, *J. Non-Cryst. Solids* 351 (24–26) 1958. doi:10.1016/j.noncrysol.05.005.
- [2] R.M. Hakim, D.R. Uhlmann, *Phys. Chem. Glasses* 8 (1967) 174.
- [3] D.E. Day, *J. Non-Cryst. Solids* 21 (1976) 343.
- [4] N.F. Mott, *Rev. Mod. Phys.* 40 (4) (1968) 677.
- [5] D. Adler, *Rev. Mod. Phys.* 40 (4) (1968) 714.
- [6] M. Imada, A. Fujimori, Y. Tokura, *Rev. Mod. Phys.* 70 (1998) 1039.
- [7] E. Mansour, G. El-Damrawi, S.A. El-Maksoud, Y.M. Moustafa, H. Doweidar, *Phys. Chem. Glasses* 43 (2) (2002) 80.
- [8] H. Doweidar, G.M. El-Damrawi, Y.M. Moustafa, *J. Phys.: Condens. Matter* 6 (1994) 8829.
- [9] B. Dutta, N.A. Fahmy, I.L. Pegg, *J. Non-Cryst. Solids*, submitted for publication.
- [10] N.F. Mott, *Conduction in Non-Crystalline Materials*, 2nd Ed., Clarendon Press, Oxford, 1993.
- [11] N.F. Mott, E.A. Davis, *Electronic Processes in Non-Crystalline Materials*, Clarendon Press, Oxford, 1979.
- [12] A.S. Alexandrov, N.F. Mott, *Polarons and Bipolarons*, World Scientific, Singapore, 1995.
- [13] N.F. Mott, *Contemp. Phys.* 14 (5) (1973) 401.
- [14] A.H. Wilson, *Proc. Roy. Soc. (London)* A133 (1931) 458.
- [15] N.F. Mott, *Proc. Phys. Soc. (London)* A62 (1949) 416.
- [16] N.F. Mott, *Phil. Mag.* 6 (1961) 287.
- [17] J. Hubbard, *Proc. R. Soc.* 276 (1963) 238.
- [18] P.W. Anderson, *Phys. Rev.* 109 (1958) 1492.
- [19] I.G. Austin, N.F. Mott, *Adv. Phys.* 18 (1969) 41.
- [20] T.E. Barnard, N.A. Fahmy, private communication.
- [21] B. Santic, A. Mogus-Milankovic, D.E. Day, *J. Non-Cryst. Solids* 296 (2001) 65.
- [22] I.A. Gohar, Y.M. Moustafa, A.A. Megahed, E. Mansour, *Phys. Chem. Glasses* 39 (1998) 56.
- [23] I.A. Gohar, Y.M. Moustafa, A.A. Megahed, E. Mansour, *Phys. Chem. Glasses* 38 (1997) 37.
- [24] N.F. Mott, *J. Non-Cryst. Solids* 1 (1968) 1.
- [25] G.N. Greaves, *J. Non-Cryst. Solids* 11 (1973) 427.
- [26] A. Lanzara, N.L. Saini, M. Brunelli, F. Natali, A. Bianconi, P.G. Radaelli, S.W. Cheong, *Phys. Rev. Lett.* 81 (1998) 878.
- [27] M. Capone, S. Ciuchi, *Phys. Rev. Lett.* 91 (2003) 186405.
- [28] T. Holstein, *Ann. Phys.* 8 (1959) 343.
- [29] D. Meyer, A.C. Hewson, R. Bulla, *Phys. Rev. B* 89 (2002) 196401.
- [30] F.X. Bronold, A. Alvermann, H. Hehske, *Phil. Mag.* 84 (7) (2004) 673.
- [31] D. Emin, *Phys. Rev. B* 48 (1993) 13691.
- [32] A.S. Alexandrov, A.M. Bratkovsky, *J. Phys.: Condens. Matter* 11 (1999) L531.